

Assignment – 1

Remote Procedure Call

1.To implement a Server Calculator using RPC concept(Make use of Server Socket)

Source Code:

CalcServer.java

```
import java.io.*;
import java.net.*;

public class CalServer {
    public static void main(String args[]) {
        System.out.println("Server is booting up...");
        try (ServerSocket serverSocket = new ServerSocket(3000)) {
            System.out.println("Waiting for a client to connect...");

            try (Socket clientsocket = serverSocket.accept()) {
                System.out.println("Client successfully connected!");

                DataInputStream cis = new
DataInputStream(clientsocket.getInputStream());
                DataOutputStream cos = new
DataOutputStream(clientsocket.getOutputStream());

                // Send initial greeting to the client
                cos.writeUTF("CONNECTED TO CALCULATOR SERVER");

                // Read input parameters from client
                int num1 = cis.readInt();
                String operator = cis.readUTF();
                int num2 = cis.readInt();

                System.out.println("Received equation: " + num1 + " " + operator + "
" + num2);
                String result;
                switch (operator) {
                    case "+": result = String.valueOf(num1 + num2); break;
```

```
        case "-": result = String.valueOf(num1 - num2); break;
        case "*": result = String.valueOf(num1 * num2); break;
        case "/":
            if (num2 == 0) {
                result = "Error: Division by zero!";
            } else {
                result = String.valueOf((double) num1 / num2);
            }
            break;
        default: result = "Error: Invalid Operator!";
    }

    cos.writeUTF(result);

    System.out.println("Sent response: " + result);
}
} catch (IOException e) {
    System.out.println("Server connection exception: " + e.getMessage());
}
}
```

CalcClient.java

```
import java.io.*;
import java.net.*;
import java.util.Scanner;

public class CalClient {
    public static void main(String args[]) {
        try (Socket socket = new Socket("localhost", 3000);
            Scanner scanner = new Scanner(System.in)) {

            DataInputStream cis = new DataInputStream(socket.getInputStream());
            DataOutputStream cos = new
DataOutputStream(socket.getOutputStream());

            System.out.println("Server says: " + cis.readUTF());

            System.out.print("Enter first integer: ");
```

```
int num1 = scanner.nextInt();

System.out.print("Enter operator (+, -, *, /): ");
String operator = scanner.next();

System.out.print("Enter second integer: ");
int num2 = scanner.nextInt();

cos.writeInt(num1);
cos.writeUTF(operator);
cos.writeInt(num2);

String serverResponse = cis.readUTF();
System.out.println("Calculator Result: " + serverResponse);

} catch (UnknownHostException e) {
    System.out.println("Server host not found: " + e.getMessage());
} catch (IOException e) {
    System.out.println("I/O exception error: " + e.getMessage());
}
}
```

Output:

```
PS S:\DSCC> javac CalClient.java
PS S:\DSCC> java CalClient
Server says: CONNECTED TO CALCULATOR SERVER
Enter first integer: 85
Enter operator (+, -, *, /): +
Enter second integer: 20
Calculator Result: 105
```

```
PS S:\DSCC> javac CalServer.java
PS S:\DSCC> java CalServer
Server is booting up...
Waiting for a client to connect...
Client successfully connected!
Received equation: 85 + 20
Sent response: 105
```

```
PS S:\DSCC> javac CalServer.java
PS S:\DSCC> java CalServer
Server is booting up...
Waiting for a client to connect...
Client successfully connected!
Received equation: 25 * 25
Sent response: 625
```

```
PS S:\DSCC> javac CalClient.java
PS S:\DSCC> java CalClient
Server says: CONNECTED TO CALCULATOR SERVER
Enter first integer: 25
Enter operator (+, -, *, /): *
Enter second integer: 25
Calculator Result: 625
```

```
PS S:\DSCC> javac CalClient.java
PS S:\DSCC> java CalClient
Server says: CONNECTED TO CALCULATOR SERVER
Enter first integer: 5
Enter operator (+, -, *, /): /
Enter second integer: 0
Calculator Result: Error: Division by zero!
```

```
PS S:\DSCC> javac CalServer.java
PS S:\DSCC> java CalServer
Server is booting up...
Waiting for a client to connect...
Client successfully connected!
Received equation: 5 / 0
Sent response: Error: Division by zero!
```

2. To implement a Date Time Server using RPC concept (Make use of Server Socket)

Source Code:

```
DateTimeServer.java
import java.io.*;
import java.net.*;
import java.time.LocalDate;
import java.time.LocalDateTime;
import java.time.LocalTime;
import java.time.format.DateTimeFormatter;

public class DateTimeServer {
    public static void main(String args[]) {

        try (ServerSocket serverSocket = new ServerSocket(3001)) {

            System.out.println("Waiting for a client to connect...");

            try (Socket clientsocket = serverSocket.accept()) {

                System.out.println("Client successfully connected!");

                DataInputStream cis =
                    new DataInputStream(clientsocket.getInputStream());

                DataOutputStream cos =
                    new DataOutputStream(clientsocket.getOutputStream());

                cos.writeUTF("CONNECTED TO DATE TIME SERVER");

                String request = cis.readUTF();

                System.out.println("Received request: " + request);

                String result;
                switch (request) {

                    case "DATE":
                        result = LocalDate.now().format(
```

```
        DateTimeFormatter.ofPattern("dd-MM-yyyy"));
        break;
    case "TIME":
        result = LocalDateTime.now().format(
            DateTimeFormatter.ofPattern("HH:mm:ss"));
        break;

    case "DATETIME":
        result = LocalDateTime.now().format(
            DateTimeFormatter.ofPattern(
                "dd-MM-yyyy HH:mm:ss"));
        break;
    default:
        result = "Error: Invalid Request!";
    }
    cos.writeUTF(result);

    System.out.println("Sent response: " + result);
}

} catch (IOException e) {
    System.out.println(
        "Server connection exception: " + e.getMessage());
}
}
```

DateTimeClient.java

```
import java.io.*;
import java.net.*;
import java.util.Scanner;
```

```
public class DateTimeClient {
    public static void main(String args[]) {

        try (Socket socket = new Socket("localhost", 3001);
            Scanner scanner = new Scanner(System.in)) {

            DataInputStream cis =
                new DataInputStream(socket.getInputStream());
```

```
DataOutputStream cos =
    new DataOutputStream(socket.getOutputStream());

System.out.println("Server says: " + cis.readUTF());

System.out.println("\nSelect an option:");
System.out.println("1. Get Date");
System.out.println("2. Get Time");
System.out.println("3. Get Date and Time");

System.out.print("Enter your choice: ");
int choice = scanner.nextInt();

String request;
switch (choice) {

    case 1:
        request = "DATE";
        break;

    case 2:
        request = "TIME";
        break;

    case 3:
        request = "DATETIME";
        break;
    default:
        request = "INVALID";
}
cos.writeUTF(request);

String serverResponse = cis.readUTF();
System.out.println("Server Response: " + serverResponse);

} catch (UnknownHostException e) {

    System.out.println(
        "Server host not found: " + e.getMessage());
```

```
    } catch (IOException e) {  
  
        System.out.println(  
            "I/O exception error: " + e.getMessage());  
    }  
}  
}
```

Output:

```
PS C:\DSCC> javac DateTimeServer.java  
PS C:\DSCC> java DateTimeServer  
Waiting for a client to connect...  
Client successfully connected!  
Received request: DATE  
Sent response: 10-08-2026
```

```
PS C:\DSCC> javac DateTimeClient.java  
PS C:\DSCC> java DateTimeClient  
Server says: CONNECTED TO DATE TIME SERVER  
  
Select an option:  
1. Get Date  
2. Get Time  
3. Get Date and Time  
Enter your choice: 1  
Server Response: 10-08-2026
```

```
PS C:\DSCC> javac DateTimeServer.java  
PS C:\DSCC> java DateTimeServer  
Waiting for a client to connect...  
Client successfully connected!  
Received request: TIME  
Sent response: 21:53:41
```

```
PS C:\DSCC> javac DateTimeClient.java
PS C:\DSCC> java DateTimeClient
Server says: CONNECTED TO DATE TIME SERVER

Select an option:
1. Get Date
2. Get Time
3. Get Date and Time
Enter your choice: 2
Server Response: 21:53:41
```

```
PS C:\DSCC> javac DateTimeServer.java
PS C:\DSCC> java DateTimeServer
Waiting for a client to connect...
Client successfully connected!
Received request: DATETIME
Sent response: 10-08-2026 21:55:40
```

```
PS C:\DSCC> javac DateTimeClient.java
PS C:\DSCC> java DateTimeClient
Server says: CONNECTED TO DATE TIME SERVER

Select an option:
1. Get Date
2. Get Time
3. Get Date and Time
Enter your choice: 3
Server Response: 10-08-2026 21:55:40
```